

PROFESSIONAL PROFILE

MEng Electronic and Electrical Engineering student at UCL looking for hardware internships in embedded systems, FPGA and digital design, PCB layout, power electronics, and robotics.

EDUCATION

University College London2024 – 2028

MEng, Electronic and Electrical Engineering

- Coursework highlights: Digital Design (78%), Advanced Digital Design (Year 3), Photonics and Communication Systems (70%), Programming and Control Systems (75%), Digital Signal Processing (Year 3).
- 3rd-Year MEng Project: Hardware-Accelerated Limit Order Book (UCL, 2026–2027).

Strode’s College2022 – 2024

- A-Levels: Mathematics (A*), Physics (A), Further Mathematics (A). GCSEs: eight grades 9 to 7, including English Language (9) and Mathematics (8).

TECHNICAL SKILLS

- Languages and HDL:** SystemVerilog, Embedded C, C/C++, Python, MATLAB.
- Hardware and Electronics:** STM32 Microcontrollers, ADC, OLED, PCB Design, SMD Soldering, FPGA Logic.
- Tools and Software:** Git, Linux/VPS, Azure Static Web Apps, CMake.

TECHNICAL EXPERIENCE

3rd-Year MEng Project: Hardware-Accelerated Limit Order Book — UCL2026 – 2027

- Designing and implementing a low-latency SystemVerilog engine on an FPGA to parse NASDAQ ITCH protocol market data streams.
- Reconstructing and maintaining deterministic limit order book state directly in hardware at wire speed.

Co-Founder and Lead Engineer — Toolbox Maths (EdTech startup backed by University of Southampton funding)2024 – Present

- Co-founded an EdTech startup building an interactive GCSE and A-Level maths platform on Vue 3, with KaTeX rendering, custom notes and whiteboard workspaces, and a progressive web app client.
- Migrating production systems onto Azure Static Web Apps, serverless Azure Functions, and Cosmos DB, with Entra authentication and Blob-backed notebook storage.
- Owned reliability work including Workbox offline caching and staged cloud deployments so students keep access during releases.

Embedded STM32 Console and Pong Game — UCL2025

- Built a handheld C/C++ game console on an STM32 that plays real-time Pong from joystick input.
- Configured STM32 ADC peripherals for continuous joystick sampling and drove an OLED display for frame updates, tuning refresh so controls stayed responsive.

Project Lead, Digital Power Supply — UCL, five-person team2025

- Led design, manufacture, and test of a custom PCB power supply converting 4 V into stable 5 V and 12 V DC outputs.
- Routed power and ground around discrete voltage regulators with thermal vias, reaching 92 percent efficiency under load and passing thermal checks.

Autonomous Maze-Solving Robot — UCL2025

- Built an autonomous maze robot with PID feedback control and ambient-light thresholding for reliable navigation across changing lighting.
- Tuned controller gains against measured steering response times and cut maze completion time by about 20 percent.

ACTIVITIES

- FPGA and digital design:** SystemVerilog modules and testbenches for small digital datapaths.
- UCL Fintech Society:** Workshops on markets and algorithmic trading.